

Uninitialized Memory Disclosures in Web Applications

Uninitialized Memory Disclosures in Web Applications - @SilentSignalHU, Silent Signal Techblog.

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Content

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```
0000:9740 0D 62 64 31 F6 00 00 C8 D6 F5 46 2C 06 00 F9 05 YD10..E00F,..U.
0000:9750 F6 06 FAD1 48 F8 7B 52 C6 BE F4 31 D4 44 66 AC ö.úNHø{RÆ3ô10Df~
0000:9760 EF 61 6D 00 65 3D 22 70 61 73 73 77 6F 72 64 22 iam_e="password"
0000:9770 0D 0A 0D 0A 50 61 73 73 77 6F 72 64 31 0D 0A 2D ... Password1 ..
0000:9780 2D 64 62 33 61 34 39 36 34 38 61 39 38 31 31 32 -db3a49648a98112
0000:9790 33 35 63 30 32 31 64 35 61 64 65 62 30 66 34 33 35c021d5adeb0f43
0000:97A0 66 0D 0A 43 6F 6E 74 65 6E 74 2D 44 69 73 70 6F f..Content-Dispo
0000:97B0 73 69 74 69 6F 6E 3A 20 66 6F 72 6D 2D 64 61 74 sition: form-dat
0000:97C0 61 3B 20 6E 61 6D 65 3D 22 65 78 74 22 0D 0A 0D a; name="ext"...
0000:97D0 0A 70 6E 67 0D 0A 2D 2D 64 62 33 61 34 39 36 34 .png...-db3a4964
0000:97E0 38 61 39 38 31 31 32 33 35 63 30 32 31 64 35 61 8a9811235c021d5a
0000:97F0 64 65 62 30 66 34 33 66 0D 0A 43 6F 6E 74 65 6E deb0f43f..Conten
0000:9800 74 2D 44 69 73 70 6F 73 69 74 69 6F 6E 3A 20 66 t-Disposition: f
0000:9810 6F 72 6D 2D 64 61 74 61 3B 20 6E 61 6D 65 3D 22 orm-data; name="
0000:9820 66 69 6C 65 22 3B 20 66 69 6C 65 6E 61 6D 65 3D file"; filename=
0000:9830 22 31 78 31 2E 70 6E 67 22 0D 0A 0D 0A 89 50 4E "1x1.png".....PN
0000:9840 47 0D 0A 1A 0A 00 00 00 0D 49 48 44 52 00 00 00 G.....IHDR...
0000:9850 01 00 00 00 01 00 00 00 00 00 00 77 53 0F 00 00 uCB
```

While we at Silent Signal are strong believers in human creativity when it comes to finding new, or unusual vulnerabilities, we're also constantly looking for ways to transform our experience into automated tools that can reliably and efficiently detect already known bug classes. The discovery of [CVE-2019-6976](#) – an uninitialized memory disclosure bug in a widely used imaging library – was a particularly interesting finding to me, as it represented a lesser known class of issues in the intersection of web application and memory safety bugs, so it seemed to be a nice topic for my next GWAPT Gold Paper.

While we did some work on investigating the issue, and even developed tooling for detection, writing a paper was a good opportunity to systematize my knowledge, and to properly evaluate the effectiveness of available discovery methods. While going through a

process where I had to back all my claims with references and data, a couple of important things quickly became apparent:

- There isn't really a standard way to think about memory safety. While [Matt Miller's work in this area](#) fit my case really well, most papers and writeups just rely on "folklore knowledge", and I realized that this makes it really hard to logically reason about one's own way of thinking (even if a particular sentence makes sense at all?).
- Some concepts that we throw around in IT-security can be [much more complex](#) than most of us probably think.
- Our original detection algorithm was suboptimal, and the existing implementation was incorrect...

Fortunately, I managed to fix the problems, and now the tools I created are available for you to verify. Following the [Unix philosophy](#) of creating simple tools that can interact once again helped me to test and compare different ideas in a reproducible and automated way. The relevant code repositories for this research are:

- [TestEnvForEntropyCalc:multi](#) – Improved branch of our Docker test environment with Apache/PHP, Node.js and Python based test applications. You can use this to experiment with new and existing tools.
- [image-memleak](#) – Test scripts referenced in the paper.
- [image-memleak-testsuite](#) – **Test images to facilitate testing of memory disclosures.** PNG's for now.

As you will see in the paper, detection of memory disclosures can be facilitated by using an appropriately chosen input test suite. Feel free to use the last repo in your tests, and if you feel like messing around with image formats, don't hesitate to contribute more samples!

I also reached out to [Chris Evans](#), whose work in this area was the original inspiration for the initial bug, and this paper too. He was kind enough to give feedback on my paper, some of which didn't make it to the released version because of timing issues:

- [Cloudbleed](#) definitely would've worth a mention in the historical overview
- Feeding the same input to the parser multiple times and looking for differences in the output seems also like a reliable way to detect parsing problems. This technique can be particularly useful, when the tested edge case doesn't allow full control over the actual bitmap content of the input image.
- As the paper mentions, it's not just image parsers that can be abused this way, these are just the most common examples one can encounter on the web. [This bug of Chris](#) is a nice example of memory disclosure in Flash (still not completely [dead](#) as of this writing).

Finally, writing this paper highlighted some areas which would deserve their own papers, such as:

- Recovering memory content after lossy compression
- Improving pointer identification based on specifics of particular executable loaders

I hope that this paper will serve as a useful foundation to better understand this exciting branch of vulnerabilities, and inspire further research. The full Gold Paper can be downloaded from the website of SANS Institute:

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